

1. Mini BA PROJECT

Clinic Booking System

Purpose of the system:

- Patients can book appointments from home
- The healthcare workers can also book surgery rooms and equipment through the system
- Reminds the user not to miss their appointment
- Provide online consultations
- Patients can order medication
- Managers can also order medicine and equipment they need

Target Users:

- IT managers and staff
- Nurses and Doctors
- Clinic Managers and Administrators
- Vendors

Introduction

The purpose of this document is to outline the business requirements of the clinic booking system. This system is going to allow users to book appointments and consultations online to reduce waiting room times and frustrations.

Objectives

The system is going to help with the following:

- Patients can book appointments from home
- The healthcare workers can also book surgery rooms and equipment through the system
- Remind the user not to miss their appointment
- Provide online consultations
- Patients can order medication
- Managers can also order medicine and equipment they need
- Reduces long queues

In scope

- User registration and login
- Booking appointments
- Booking surgery rooms and equipment
- Viewing available time slots
- Appointment confirmation

Out of scope

- Online payments
- Integration with external systems

Stakeholders

- IT
- Patients
- Nurses
- Doctors
- Clinic Managers
- Administration
- Vendors
- Clinic staff

Functional Requirements

- User registers an account
- User logs in
- User views available time slots
- User books appointment
- User receives appointment confirmation

Non-functional Requirements

- The system must load within 4 seconds
- The system must be available 24/7
- The system must ensure data privacy and security
- The system must be user-friendly

User Stories

- As a patient, I want to register an account so that I can book appointments
- As a patient, I want to log in and access my bookings
- As a patient, I want to book an appointment so I can see a doctor
- As a patient, I want to receive confirmation for my booking

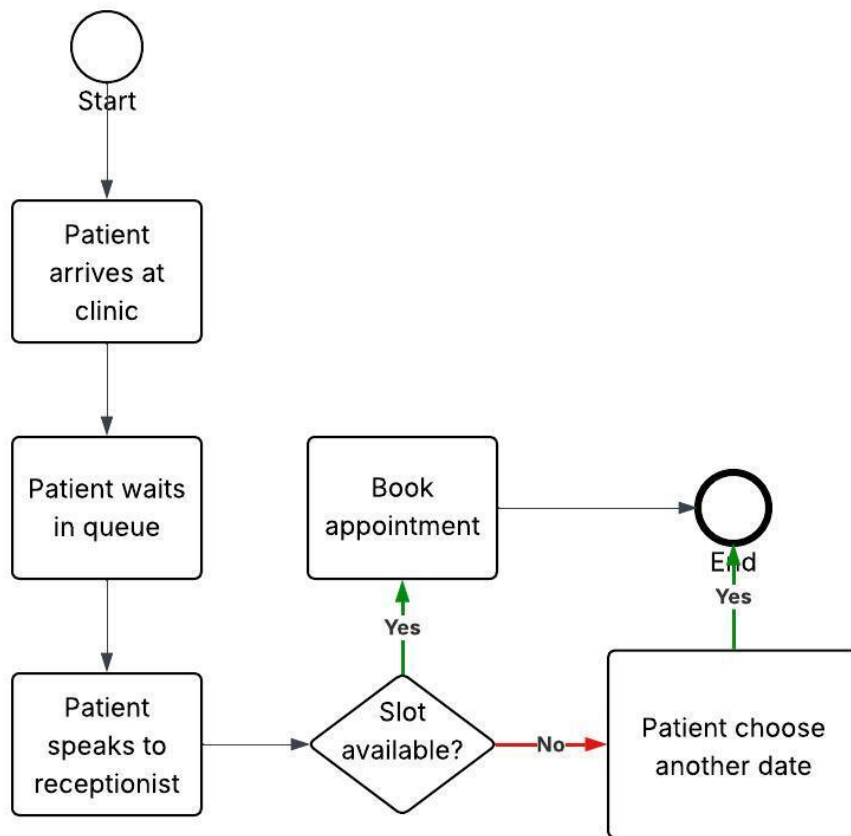
The system will be considered successful if:

- Users can register and log in successfully
- Users can book appointments without errors
- The system displays available slots correctly

Process Flow

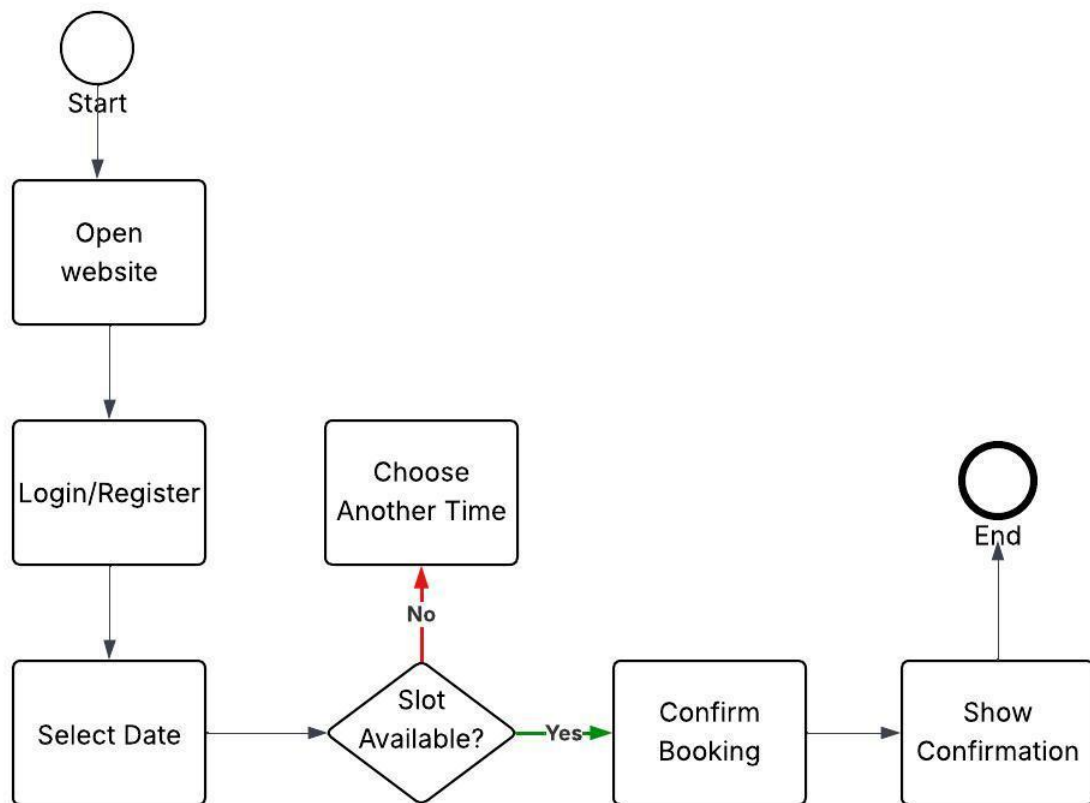
Current Manual System (AS-IS)

- Patient arrives at the clinic
- Patient waits in queue
- Patient speaks to receptionist
- Receptionist checks availability
- Appointment is written in a book
- Patient is given a time slot



New System Process (TO-BE)

- User goes to website
- User register account
- User logs in
- User checks availability
- User books and appointment
- User receives appointment confirmation

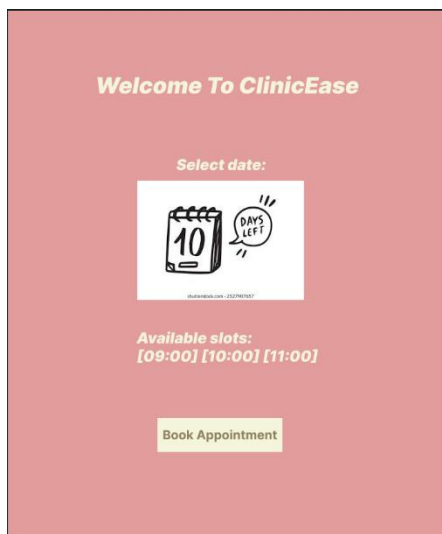


System Screens

Screen 1: Log in Page

The image shows a mobile app login screen for 'ClinicEase'. The header is purple with the ClinicEase logo and tagline 'Smart bookings, seamless care'. The main area has a pink background and contains input fields for 'Email Address' and 'Password', a 'Log In' button, and a link 'Don't have an account? Register'.

Screen 2: Booking Page



A mockup of a booking page with a light red background. At the top, it says "Welcome To ClinicEase" in a bold, sans-serif font. Below this, there is a "Select date:" label followed by a calendar icon showing the number 10 and a speech bubble that says "DAYS LEFT!". Underneath the calendar, it says "Available slots:" followed by three time slots: "[09:00] [10:00] [11:00]". At the bottom, there is a yellow button labeled "Book Appointment".

Welcome To ClinicEase

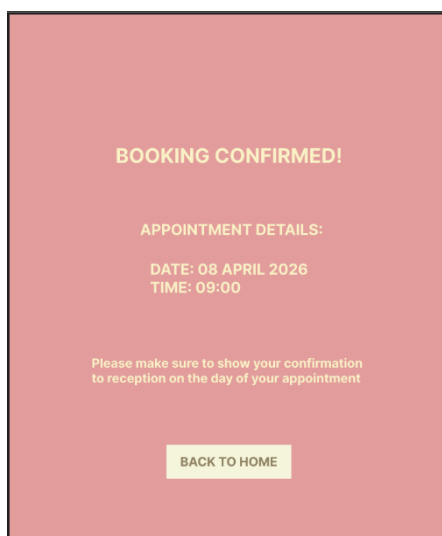
Select date:

10 DAYS LEFT!

Available slots:
[09:00] [10:00] [11:00]

Book Appointment

Screen 3: Confirmation Page



A mockup of a confirmation page with a light red background. At the top, it says "BOOKING CONFIRMED!" in a bold, sans-serif font. Below this, there is a label "APPOINTMENT DETAILS:" followed by the date "DATE: 08 APRIL 2026" and the time "TIME: 09:00". Underneath, there is a paragraph of text: "Please make sure to show your confirmation to reception on the day of your appointment". At the bottom, there is a yellow button labeled "BACK TO HOME".

BOOKING CONFIRMED!

APPOINTMENT DETAILS:

DATE: 08 APRIL 2026
TIME: 09:00

Please make sure to show your confirmation
to reception on the day of your appointment

BACK TO HOME

To access the prototype and see how the system works, click on the link:

<https://www.figma.com/proto/Id5ucSUXTHElaTLbgO1N6A/Untitled?node-id=1-2&starting-point-node-id=1%3A2&t=bUfNciKa2TdkoA5w-1>

2. Clinic Data Analysis Using SQL

Project Description

In this project, I used SQL to analyse clinic data, including patient activity, appointments, and revenue. I performed data aggregation, joins across multiple tables, and generated business insights to support decision-making.

Database Structure

Tables Used:

- Patients
- Appointments
- Payments

These tables store patient information, appointment details, and payment transactions.

Skills Demonstrated

- SQL Queries (SELECT, WHERE, GROUP BY)
- JOIN operations (INNER JOIN)
- Data aggregation (SUM, COUNT)
- Business insight generation

DATABASE STRUCTURE

Create these tables in a database using SQL:

- **Patients Table**

```
CREATE TABLE Patients (  
    Patient_ID INT,  
    Name VARCHAR (50),  
    City VARCHAR (50)  
);
```

- **Appointments Table**

```
CREATE TABLE Appointments (
```

```
Appointment_ID INT,  
Patient_ID INT,  
Date DATE,  
Service VARCHAR (50)  
);
```

- **Payments Table**

```
CREATE TABLE Payments (  
Payment_ID INT,  
Appointment_ID INT,  
Amount DECIMAL (10.2)  
);
```

Insert sample data:

- INSERT INTO Patients VALUES
(1, 'John', 'London'),
(2, 'Sarah', 'Paris'),
(3, 'Mike', 'London');
- INSERT INTO Appointments VALUES
(1, 1, '2026-03-01', 'General Checkup'),
(2, 2, '2026-03-02', 'Dental'),
(3, 1, '2026-03-03', 'Eye Test'),
(4, 3, '2026-03-03', 'General Checkup');
- INSERT INTO Payments VALUES
(1, 1, 500),
(2, 2, 300),
(3, 3, 400),
(4, 4, 500);

1. What is the total money earned?

```
SELECT SUM(amount) AS total_revenue  
FROM Payments;
```


Schema SQL

```
1 CREATE TABLE Patients (  
2   Patient_ID INT,  
3   Name VARCHAR(50),  
4   City VARCHAR(50)  
5 );  
6  
7 CREATE TABLE Appointments (  
8   Appointment_ID INT,  
9   Patient_ID INT,  
10  Date DATE,  
11  Service VARCHAR(50)  
12 );  
13  
14 CREATE TABLE Payments (  
15   Payment_ID INT,  
16   Appointment_ID INT,  
17   Amount DECIMAL(10,2)  
18 );  
19  
20 TRUNCATE TABLE Patients, Appointments, Payments;
```

Text to DDL

Query SQL

```
1 SELECT SUM(amount) AS total_revenue  
2 FROM Payments;
```

Results

Query #1 Execution time: 0.18ms

total_revenue
1700.0

The total money earned is 1700.00

2. Who has the most appointments?

```
SELECT Patient_ID, COUNT(*) AS Total_Appointments  
FROM Appointments  
GROUP BY Patient_ID  
ORDER BY Total_Appointments DESC;
```

Schema SQL

```
1 CREATE TABLE Patients (  
2   Patient_ID INT,  
3   Name VARCHAR(50),  
4   City VARCHAR(50)  
5 );  
6  
7 CREATE TABLE Appointments (  
8   Appointment_ID INT,  
9   Patient_ID INT,  
10  Date DATE,  
11  Service VARCHAR(50)  
12 );  
13  
14 CREATE TABLE Payments (  
15   Payment_ID INT,  
16   Appointment_ID INT,  
17   Amount DECIMAL(10,2)  
18 );  
19  
20 TRUNCATE TABLE Patients, Appointments, Payments;
```

Text to DDL

Query SQL

```
1  
2 SELECT Patient_ID, COUNT(*) AS total_appointments  
3 FROM Appointments  
4 GROUP BY Patient_ID  
5 ORDER BY total_appointments DESC;
```

Results

Query #1 Execution time: 0.41ms

Patient_ID	total_appointments
1	2
2	1
3	1

Patient John visits the clinic most frequently

3. Which service is booked the most?

```
SELECT Service, COUNT(*) AS Total  
FROM Appointments
```

GROUP BY Service ORDER BY Total DESC;

Schema SQL

```
1 CREATE TABLE Patients (  
2   Patient_ID INT,  
3   Name VARCHAR(50),  
4   City VARCHAR(50)  
5 );  
6  
7 CREATE TABLE Appointments (  
8   Appointment_ID INT,  
9   Patient_ID INT,  
10  Date DATE,  
11  Service VARCHAR(50)  
12 );  
13  
14 CREATE TABLE Payments (  
15   Payment_ID INT,  
16   Appointment_ID INT,  
17   Amount DECIMAL(10,2)  
18 );  
19  
20 TRUNCATE TABLE Patients, Appointments, Payments;
```

Text to DDL

Query SQL

```
1 SELECT Service, COUNT(*) AS Total  
2 FROM Appointments  
3 GROUP BY Service  
4 ORDER BY Total DESC
```

Results

Query #1 Execution time: 0.33ms

Service	Total
GENERAL CHECKUP	2
DENTAL	1
EYE TEST	1

General checkups are booked the most, therefore it generates the highest revenue.

4. Join all tables

```
SELECT Patients.Name, Appointments.Service, Payments.Amount  
FROM Patients  
INNER JOIN Appointments  
ON Patients.Patient_ID = Appointments.Patient_ID  
INNER JOIN Payments  
ON Appointments.Appointment_ID = Payments.Appointment_ID;
```

Schema SQL

```
1 CREATE TABLE Patients (  
2   Patient_ID INT,  
3   Name VARCHAR(50),  
4   City VARCHAR(50)  
5 );  
6  
7 CREATE TABLE Appointments (  
8   Appointment_ID INT,  
9   Patient_ID INT,  
10  Date DATE,  
11  Service VARCHAR(50)  
12 );  
13  
14 CREATE TABLE Payments (  
15   Payment_ID INT,  
16   Appointment_ID INT,  
17   Amount DECIMAL(10,2)  
18 );  
19  
20 TRUNCATE TABLE Patients, Appointments, Payments;
```

Text to DDL

Query SQL

```
1 SELECT Patients.Name, Appointments.Service, Payments.Amount  
2 FROM Patients  
3 INNER JOIN Appointments  
4 ON Patients.Patient_ID = Appointments.Patient_ID  
5 INNER JOIN Payments  
6 ON Appointments.Appointment_ID = Payments.Appointment_ID
```

Results

Query #1 Execution time: 0.25ms

Name	Service	Amount
JOHN	GENERAL CHECKUP	500.0
SARAH	DENTAL	300.0
JOHN	EYE TEST	400.0
MIKE	GENERAL CHECKUP	500.0

5. How much does each service generate?

```
SELECT Appointments.Service, SUM(Payments.Amount) AS Revenue  
FROM Appointments
```

INNER JOIN Payments

ON Appointments.Appointment_ID = Payments.Appointment_ID

GROUP BY Appointments.Service;

Schema SQL

```
1 CREATE TABLE Patients (  
2   Patient_ID INT,  
3   Name VARCHAR(50),  
4   City VARCHAR(50)  
5 );  
6  
7 CREATE TABLE Appointments (  
8   Appointment_ID INT,  
9   Patient_ID INT,  
10  Date DATE,  
11  Service VARCHAR(50)  
12 );  
13  
14 CREATE TABLE Payments (  
15   Payment_ID INT,  
16   Appointment_ID INT,  
17   Amount DECIMAL(10,2)  
18 );  
19  
20 TRUNCATE TABLE Patients, Appointments, Payments;
```

Text to DDL

Query SQL

```
1 SELECT Appointments.Service, SUM(Payments.Amount) AS Revenue  
2 FROM Appointments  
3 INNER JOIN Payments  
4 ON Appointments.Appointment_ID = Payments.Appointment_ID  
5 GROUP BY Appointments.Service;
```

Results

Query #1 Execution time: 0.34ms

Service	Revenue
DENTAL	300.0
EYE TEST	400.0
GENERAL CHECKUP	1000.0

General checkups generate the highest revenue

6. Find patients who made more than 1 appointment

SELECT Patient_ID

FROM Appointments

GROUP BY Patient_ID

HAVING COUNT(*) > 1;

Schema SQL

```
1 CREATE TABLE Patients (  
2   Patient_ID INT,  
3   Name VARCHAR(50),  
4   City VARCHAR(50)  
5 );  
6  
7 CREATE TABLE Appointments (  
8   Appointment_ID INT,  
9   Patient_ID INT,  
10  Date DATE,  
11  Service VARCHAR(50)  
12 );  
13  
14 CREATE TABLE Payments (  
15   Payment_ID INT,  
16   Appointment_ID INT,  
17   Amount DECIMAL(10,2)  
18 );  
19  
20 TRUNCATE TABLE Patients, Appointments, Payments;
```

Text to DDL

Query SQL

```
1 SELECT Patient_ID  
2 FROM Appointments  
3 GROUP BY Patient_ID  
4 HAVING COUNT(*) > 1
```

Results

Query #1 Execution time: 0.25ms

Patient_ID
1

There was only 1 patient who made more 1 appointment

7. Get names of frequent patients

SELECT Name

FROM Patients

WHERE Patients_ID IN (

SELECT Patient_ID

```

FROM Appointments
GROUP BY Patient_ID
HAVING COUNT(*) > 1
);

```

Schema SQL

```

1 CREATE TABLE Patients (
2   Patient_ID INT,
3   Name VARCHAR(50),
4   City VARCHAR(50)
5 );
6
7 CREATE TABLE Appointments (
8   Appointment_ID INT,
9   Patient_ID INT,
10  Date DATE,
11  Service VARCHAR(50)
12 );
13
14 CREATE TABLE Payments (
15   Payment_ID INT,
16   Appointment_ID INT,
17   Amount DECIMAL(10,2)
18 );
19
20 TRUNCATE TABLE Patients;

```

Text to DDL

Query SQL

```

1 SELECT Name
2 FROM Patients
3 WHERE Patient_ID IN (
4   SELECT Patient_ID
5   FROM Appointments
6   GROUP BY Patient_ID
7   HAVING COUNT(*) > 1
8 );

```

Results

Query #1

Execution time: 0.31ms

Name
JOHN

John visits the clinic frequently

8. Categorize payments

```

SELECT Amount,
CASE
    WHEN Amount >= 500 THEN 'High'
    WHEN Amount >= 300 THEN 'Medium'
    ELSE 'Low'
END AS Payment_Category
FROM Payments;

```

Schema SQL

```

1 CREATE TABLE Patients (
2   Patient_ID INT,
3   Name VARCHAR(50),
4   City VARCHAR(50)
5 );
6
7 CREATE TABLE Appointments (
8   Appointment_ID INT,
9   Patient_ID INT,
10  Date DATE,
11  Service VARCHAR(50)
12 );
13
14 CREATE TABLE Payments (
15   Payment_ID INT,
16   Appointment_ID INT,
17   Amount DECIMAL(10,2)
18 );
19
20 TRUNCATE TABLE Patients;

```

Text to DDL

Query SQL

```

1 SELECT Amount,
2 CASE
3   WHEN Amount >= 500 THEN 'High'
4   WHEN Amount >= 300 THEN 'Medium'
5   ELSE 'Low'
6 END AS Payment_Category
7 FROM Payments;

```

Results

Query #1

Execution time: 0.22ms

Amount	Payment_Category
500.0	High
300.0	Medium
400.0	Medium
500.0	High

9. Daily Booking Analysis

```

SELECT Date, COUNT(*) AS Total_Appointments
FROM Appointments

```

GROUP BY Date
ORDER BY Total_Appointments DESC

Schema SQL

```
1 CREATE TABLE Patients (  
2   Patient_ID INT,  
3   Name VARCHAR(50),  
4   City VARCHAR(50)  
5 );  
6  
7 CREATE TABLE Appointments (  
8   Appointment_ID INT,  
9   Patient_ID INT,  
10  Date DATE,  
11  Service VARCHAR(50)  
12 );  
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14 CREATE TABLE Payments (  
15   Payment_ID INT,  
16   Appointment_ID INT,  
17   Amount DECIMAL(10,2)  
18 );  
19  
20 TRUNCATE TABLE Patients, Appointments, Payments;
```

Text to DDL

Query SQL

```
1 SELECT Date, COUNT(*) AS Total_Appointments  
2 FROM Appointments  
3 GROUP BY Date  
4 ORDER BY Total_Appointments DESC;
```

Results

Query #1 Execution time: 0.24ms

Date	Total_Appointments
2026-03-03	2
2026-03-01	1
2026-03-02	1

10. Top Performing Service
SELECT Service, COUNT(*) AS Total
FROM Appointments
GROUP BY Service
ORDER BY Total DESC
LIMIT 1;

Schema SQL

```
1 CREATE TABLE Patients (  
2   Patient_ID INT,  
3   Name VARCHAR(50),  
4   City VARCHAR(50)  
5 );  
6  
7 CREATE TABLE Appointments (  
8   Appointment_ID INT,  
9   Patient_ID INT,  
10  Date DATE,  
11  Service VARCHAR(50)  
12 );  
13  
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15   Payment_ID INT,  
16   Appointment_ID INT,  
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```

Text to DDL

Query SQL

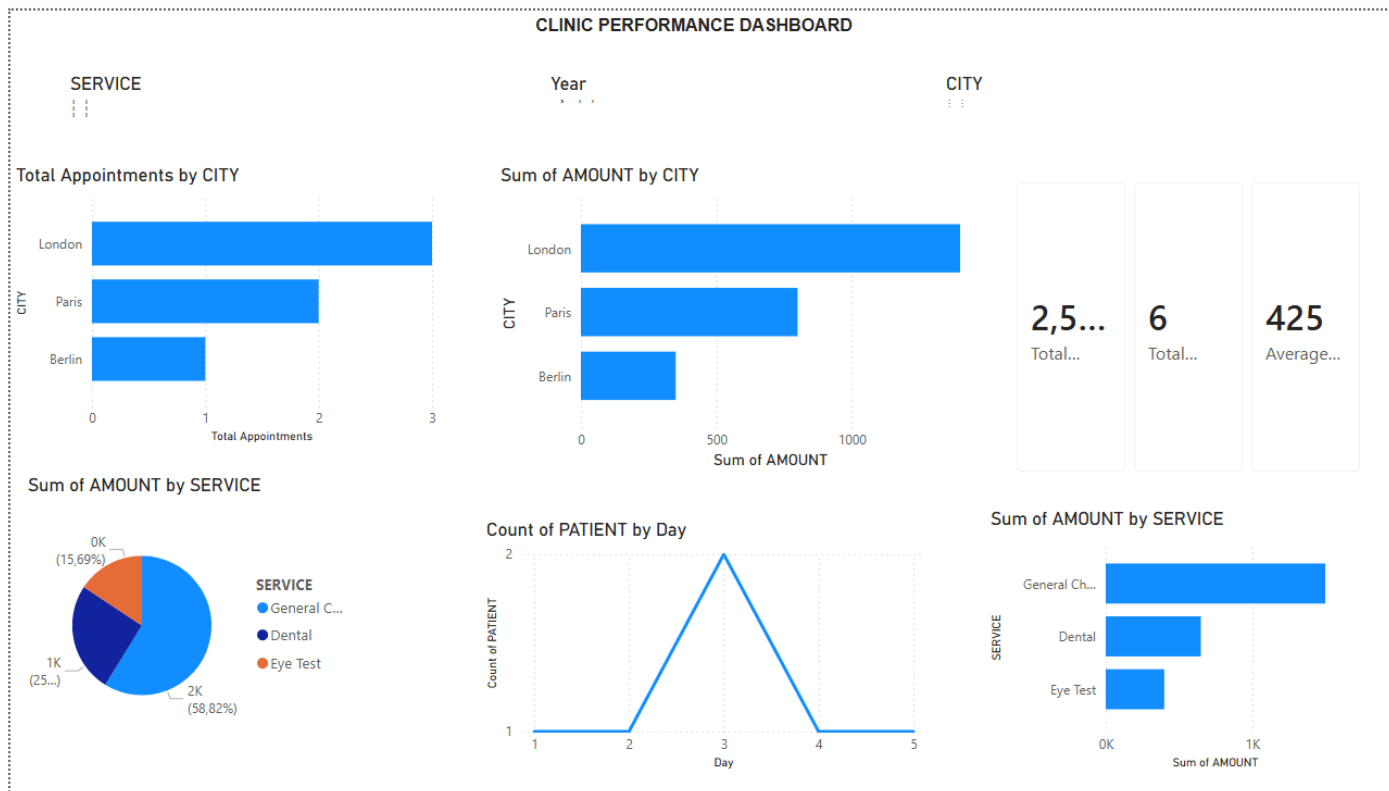
```
1 SELECT Service, COUNT(*) AS Total  
2 FROM Appointments  
3 GROUP BY Service  
4 ORDER BY Total DESC  
5 LIMIT 1;  
6
```

Results

Query #1 Execution time: 0.56ms

Service	Total
GENERAL CHECKUP	2

3. Data Visuals



1. Identify top-performing service:

The top-performing service is General Checkups, generating the highest revenue among all services. This indicates that it is the most in-demand and contributes significantly to the clinic's income.

2. Identify busiest day:

The busiest day for appointments occurs on the 3rd of March 2026, where the highest number of patient visits is recorded. This suggests peak demand during this period.

3. Compare revenue by city:

London generates the highest revenue compared to other cities, indicating stronger patient activity and higher service usage in this location.

4. Highlight trends over time:

The data shows fluctuations in appointment activity over time, with noticeable peaks and drops. This suggests that demand is not consistent and may depend on specific days or factors.

5. Recommend improvements:

The clinic should focus on promoting high-performing services such as General Checkups while implementing strategies to improve lower-performing services. Additionally, resources should be allocated efficiently during peak days to manage higher patient demand.

